

# Project 2 (EAS502 : Intro Prob Theory)

Rishi Shanthan Bhagavatham

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Submitted to : Dr. Vochita Mihai

I. For all the Question, please solve the problems using a code in R. Save the document according with the instructions provided in the attached document.

## Question 1

1). You are taking a 15-question multiple choice quiz and each question has 5 options (a, b, c, d, e) and you randomly guess every question.

- How many questions do you expect to answer correct on average?
- What is the probability that you get every question correct?
- What is the probability that you get every question incorrect?
- How many questions does one need to answer correctly in order score exactly a 60%?
- If a grade of 60% or lower is considered failing, then what is the probability of you failing?
- If you need a grade of 80% or higher on this quiz to maintain a passing grade, what is the probability of you maintaining that passing grade?

```
# Q1 a
q1a_answer <- 15 * (1 / 5)
cat("Number of questions we expect to answer correct on average:",q1a_answer)
```

```
## Number of questions we expect to answer correct on average: 3
```

```
# Q1 b
q1b_answer <- (1 / 5)^15
cat("probability that we get every question correct:",q1b_answer)
```

```
## probability that we get every question correct: 3.2768e-11
```

```
cat("Probability that we get every question correct (Using floor function):",floor(q1b_a
nswer))
```

```
## Probability that we get every question correct (Using floor function): 0
```

```
# Q1 c
q1c_answer <- ((5 - 1)/5)^15
q1c_answer * 100
```

```
## [1] 3.518437
```

```
cat("Probability that you get every question incorrect:",q1c_answer)
```

```
## Probability that you get every question incorrect: 0.03518437
```

```
# Q1 d
q1d_answer <- 15 * 0.60
cat("Questions does one need to answer correctly in order score exactly a 60% :",q1d_answer)
```

```
## Questions does one need to answer correctly in order score exactly a 60% : 9
```

```
# Q1 e
q1e_answer <- pbinom(q = q1d_answer, size = 15, prob = 0.2)
q1e_answer
```

```
## [1] 0.9998868
```

```
cat("Probability of we failing :",q1e_answer)
```

```
## Probability of we failing : 0.9998868
```

```
# Q1 f
q1f_answer <- pbinom(q = 11.85, size = 15, prob = .2, lower.tail = FALSE)
cat("Probability of us maintaining that passing grade:",q1f_answer)
```

```
## Probability of us maintaining that passing grade: 1.011253e-06
```

## Question 2

2). A fair die is rolled repeatedly. Let  $X$  be the numbers of rolls needed to obtain a 5 and  $Y$  the number of rolls needed to obtain a 6. Calculate the expected value of  $X$  given that  $Y = 2$ .

```

p <- 1/6

p_x_given_1 <- 1/5

# E(X|X=1) is just 1, since we've rolled a 5 first
E_X_given_X_1 <- 1

# So we are looking for the expected value starting from roll 3
E_X_given_X_geq_3 <- 1/p + 2 # We add 2 because we know two rolls have already occurred

# Calculate the total expected value
E_X_given_Y_2 <- E_X_given_X_1 * p_x_given_1 + E_X_given_X_geq_3 * (1 - p_x_given_1)

cat("Expected value of X given Y=2 :",E_X_given_Y_2)

```

```
## Expected value of X given Y=2 : 6.6
```

## Question 3

3). Assume IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. If a person is randomly selected, find each of the requested probabilities. Here,  $X$ , denotes the IQ of the randomly selected person.

1.  $P(X > 65)$
2.  $P(X < 150)$

```

q3_mean <- 100
q3_sd <- 15

# Q3 a
q3_a <- (1 - pnorm(65,q3_mean,q3_sd))
cat("Value of P(X > 65) :",q3_a)

```

```
## Value of P(X > 65) : 0.9901847
```

```

# Q3 b
q3_b <- pnorm(150,q3_mean,q3_sd)
cat("Value of P(X < 150) :",q3_b)

```

```
## Value of P(X < 150) : 0.9995709
```

## Question 4

4). Assume the same mean and standard deviation of IQ scores that was described in question 3.

- a. A high school offers a special program for gifted students. In order to qualify, students must have IQ scores in the top 5%. What is the minimum qualifying IQ?
- b. If one person is randomly selected, what is the probability that their IQ score is greater than 125?
- c. What is the z-score for an IQ of 140?
- d. A data value is considered “unusual” if it lies more than two standard deviations from the mean. Is an IQ of 140 considered unusual?
- e. What is the probability of getting an IQ greater than 140?

```
# Q4 a
q4_a <- qnorm(0.95, q3_mean, q3_sd)
cat("Minimum qualifying IQ :",q4_a)
```

```
## Minimum qualifying IQ : 124.6728
```

```
# Q4 b
q4_b <- (1 - pnorm(125, q3_mean, q3_sd))
cat("The IQ score is greater than 125 :",q4_b)
```

```
## The IQ score is greater than 125 : 0.04779035
```

```
# Q4 c
q4_c <- (140 - q3_mean) / q3_sd
cat("The z-score for an IQ of 140 :",q4_c)
```

```
## The z-score for an IQ of 140 : 2.666667
```

```
# Q4 d
q4_d <- abs(140 - q3_mean) > 2 * q3_sd
cat("Is an IQ of 140 considered unusual: ",q4_d)
```

```
## Is an IQ of 140 considered unusual: TRUE
```

```
# Q4 e
q4_e <- pnorm(140, q3_mean, q3_sd, lower.tail = FALSE)
q4_e
```

```
## [1] 0.003830381
```

```
cat("Probability of getting an IQ greater than 140: ",q4_e)
```

```
## Probability of getting an IQ greater than 140: 0.003830381
```

## Question 5

5). 5. Suppose you own a catering company. You hire local college students as servers. Not being the most reliable employees, there is an 80% chance that any one server will actually show up for a scheduled event. For a wedding scheduled on Saturday, you need at least 5 servers.

- Suppose you schedule 5 employees, what is the probability that all 5 come to work?
- Suppose you schedule 7 employees, what is the probability that at least 5 come to work?
- It is really important that you have at least 5 servers show up! How many employees should you schedule in order to be 99% confident that at least 5 show up?

```
# Q5 a
q5_a <- dbinom(5, 5, 4/5)
q5_a * 100
```

```
## [1] 32.768
```

```
cat("Probability that all 5 come to work :",q5_a)
```

```
## Probability that all 5 come to work : 0.32768
```

```
# Q5 b
q5_b <- 1 - pbinom(4, 7, 4/5)
q5_b * 100
```

```
## [1] 85.1968
```

```
cat("Probability that at least 5 come to work :",q5_b)
```

```
## Probability that at least 5 come to work : 0.851968
```

```
# Q5 c
for(q5_c in 5:20) {
  prob_server_show_up <- 1 - pbinom(4, q5_c, 4/5)
  if (prob_server_show_up >= 0.99) {
    break
  }
}
cat("Number of employees we can schedule in order to be 99% confident that at least 5 show up :",q5_c)
```

```
## Number of employees we can schedule in order to be 99% confident that at least 5 show up : 10
```

```
library(ggplot2)

scheduled <- 5:20

cumulative_probabilities <- 1 - pbinom(4, scheduled, 4/5)

data_frame <- data.frame(Scheduled = scheduled, Probability = cumulative_probabilities)

ggplot(data_frame, aes(x = Scheduled, y = Probability)) +
  geom_line() + # Line plot
  geom_point() + # Add points to the line plot
  geom_hline(yintercept = 0.99, linetype="dashed", color = "red") + # Line for 99% confidence
  scale_y_continuous(labels = scales::percent_format()) + # Convert y-axis to percentage
  theme_minimal() +
  labs(title = "Probability of At Least 5 Servers Showing Up",
       x = "Number of Employees Scheduled",
       y = "Probability of At Least 5 Showing Up")
```

